Probability and Statistics (SCS 1307)

Short Note

Contents

1 Introduction to Statistics and Basic Concepts

1.1 What is Statistics?

Statistics is the science of collecting, organizing, analyzing and presenting data, and finally drawing conclusions from that data in the best possible way.

1.1.1 Why We Study Statistics

- To evaluate published data.
- To collect/summarize data meaningfully.
- To present findings (in reports, or verbally).
- To interpret statistical results.
- To employ statistical methods to make decisions.
- To develop new techniques.
- To generalize decisions.

Common statistical software used in practice includes EXCEL, SPSS, MINITAB, SAS, S-Plus, MATLAB, R, and Python.

1.2 Basic Terms

- **Population**: the complete collection of individuals or objects that are of interest to study.
- **Sample**: a subset of the population.
- **Variable**: an observable, measurable or recordable characteristic; the basic unit of analysis.
- **Observation**: a value that a variable assumes for a single element.
- **Data**: the set of observations collected for a variable.
- **Parameter**: a numerical summary measure used to describe a characteristic of a *population*.
- **Statistic**: a numerical summary measure used to describe a characteristic of a *sample*.

1.3 Summarizing Data

In practice we usually have a huge amount of data, and it is difficult to look at each individual observation and draw meaningful conclusions. We therefore need ways to summarize data so as to bring out its main features.

1.4 Types of Variables and Levels of Measurement

Variables are characteristics of interest that we wish to study. Different statistical methods are appropriate for different types of variables. There are two broad kinds of variables: **Numerical (Quantitative) Variables** and **Categorical (Qualitative) Variables**.

1.4.1 Categorical (Qualitative) Variables

A variable having categories or classifications that are not numerical in nature. *Examples*: gender (male, female); social class (high, medium, low); quality of computers (good, bad); hair colour; ethnicity; whether income tax was paid last year (yes/no).

1.4.2 Numerical (Quantitative) Variables

A variable whose values are numerical in nature. *Examples*: weight, height, volume, exam marks, number of defective parts in a computer, age of a person, blood cholesterol level.

Quantitative variables can be further subdivided into **Discrete** and **Continuous** variables:

- **Discrete variable**: there is a gap between two possible values; it takes only countable or finite values. *Examples*: number of customers arriving at a supermarket; number of defective parts in a computer.
- **Continuous variable**: theoretically, there is no gap between two possible values; it takes an uncountable number of values (any real value in an interval). *Examples*: amount of rainfall; time taken to complete a computer job.

Overall classification tree:

Variables → {Numerical (Quantitative): Discrete, Continuous} or {Categorical (Qualitative): Ordinal, Nominal}

Categorical variables are further classified as **Nominal** (categories with no natural order, e.g. blood group) or **Ordinal** (categories with a natural order, e.g. social class: High/Medium/Low).

1.5 Measures of Location

- **Minimum**: the first observation of the ordered data set.
- **Maximum**: the last observation of the ordered data set.
- **Measure of Center**: a quantity that locates the center of the data set. The most common measures of center are the **Mean**, **Median** and **Mode**.
- **Quartiles**: quantities that divide the data set into four equal parts.
- **Percentiles**: quantities that divide the data set into a hundred equal parts.

1.5.1 Mean

The average value of the observations:

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \cdots + x_n}{n} = \frac{\sum_{i=1}^{n} x_i}{n}$$

The mean is very sensitive to extreme values. There are different versions of the mean:

- **Arithmetic mean** (the ordinary mean, as above).
- **Weighted mean**: used when we wish to give greater emphasis to some values. With weights $w_1, w_2, \ldots, w_n$:
$$\bar{x}_w = \frac{\sum_{i=1}^{n} w_i x_i}{\sum_{i=1}^{n} w_i}$$

1.5.2 Median (M)

The value which divides the ordered data set into two equal parts: half of the observations are smaller and the other half are larger than M.

Steps to find the median:

1. Arrange all observations in order (smallest to largest).

2. If the number of observations n is:
   - **odd**: the center value, $M = \left(\frac{n+1}{2}\right)^{\text{th}}$ value.
   - **even**: the mean of the two middle values,

$$M = \frac{\left(\frac{n}{2}\right)^{\text{th}} \text{value} + \left(\frac{n}{2}+1\right)^{\text{th}} \text{value}}{2}$$

1.5.3 Mean versus Median

- The mean is the *informal* midpoint of the data set – the data set may or may not be divided into two equal parts by it.
- The median is the *formal* midpoint of the data set – exactly at the middle, the data set is divided into two equal parts.
- The mean and median coincide only if the distribution is symmetrical.
- The median is resistant to skew and outliers; the mean is not.
- For a symmetric distribution: Mean = Median.
- For a left-skewed distribution: Mean $<$ Median.
- For a right-skewed distribution: Median $<$ Mean.

1.5.4 Mode

The most frequently occurring observation.

- There may be more than one mode (multi-modal data).
- There may be a data set with *no* mode at all (if every value occurs equally often).
- Less frequently used than the mean or median.
- The **only** measure of center that can be used for both categorical and numerical data (e.g. the modal blood group in a sample).

1.5.5 Resistant Measures

A **resistant measure** does not respond strongly to extreme values or to changes in a few observations, no matter how large those changes are.

- The **mean** is *not* a resistant measure.
- The **median** and **mode** *are* resistant measures.

1.5.6 Quartiles

Divide the ordered data into four equal subsets:

$$Q_1 = \left(\frac{1(n+1)}{4}\right)^{\text{th}} \text{value}, \qquad Q_2 = \left(\frac{2(n+1)}{4}\right)^{\text{th}} \text{value}, \qquad Q_3 = \left(\frac{3(n+1)}{4}\right)^{\text{th}} \text{value}$$

1.5.7 Percentiles

Divide the ranked data into 100 equal subsets. The k-th percentile (P_k) is the value such that $k\%$ of the data are smaller and $(100 - k)\%$ are larger.

Notes: $Q_1 = P_{25}$, $Q_3 = P_{75}$, Median $= P_{50} = Q_2$.

Procedure to obtain P_k:

1. Rank the n data values from smallest to largest.
2. Calculate $nk/100$.
3. **If** $nk/100$ **is an integer** a: let $d(P_k) = a.5$ (i.e. we will average positions a and $a + 1$).
   If $nk/100$ **has a fractional part**, giving result b (not an integer): let $d(P_k) =$ the next larger integer above b.
4. **If** $d(P_k) = a.5$: P_k is the average of the a-th and $(a + 1)$-th values.
   If $d(P_k) = b$ (an integer as computed above): P_k is the value of the data in the b-th position.

Worked example: Exam marks of 50 students are given. To find the first quartile (Q_1), the 58th percentile, and the third quartile (Q_3):

1. Rank the 50 data values from smallest to largest.
2. Compute $nk/100 = 50 \times 25/100 = 12.5$ for Q_1 (i.e. $k = 25$).
3. Since 12.5 has a fractional part, $d(Q_1) = 13$ (the next larger integer above 12.5).
4. Q_1 is the 13th value from the lowest – in this data set, $Q_1 = 67$.

(Similarly, it can be verified that $P_{58} = 77.5$ and $Q_3 = 86$ for this particular data set.)

1.6 Measures of Dispersion (Spread / Variability)

Measures of dispersion describe how far each value is spread away from the measure of center. Without a measure of dispersion, a measure of center alone can be misleading.

Illustration: the following three data sets all have median 6, but very different spreads:

Data	Median	Spread
2, 5, 6, 8, 9	6	$9 - 2 = 7$
5, 5, 6, 6, 6	6	$6 - 5 = 1$
6, 6, 6, 6, 6	6	$6 - 6 = 0$

A good measure of variability should:

- be a real number;
- be zero if all the data are identical;
- increase as the data becomes more diverse;
- never be less than zero.

1.6.1 Range and Interquartile Range

- **Range (R)**: $R =$ largest value – smallest value.
  - The simplest measure of dispersion.
  - Gives an indication of the absolute spread of the distribution.

- Depends only on the two extreme values.
- Not very informative, and strongly affected by outliers.

- **Interquartile Range (IQR)**: $IQR = Q_3 - Q_1$.
  - The range of the middle 50% of the values.
  - The upper 25% and lower 25% of observations are discarded.
  - Not affected by outliers.
  - Not necessarily centered on the middle, unless the distribution is symmetric.
  - Often used together with the median.

1.6.2 Variance

Numerically, the variance equals the average of the squared deviations from the mean. The variance for a sample of size n is:

$$S^2 = \frac{\sum(x_i - \bar{x})^2}{n-1} = \frac{\sum x_i^2 - n\bar{x}^2}{n-1}$$

A data set which is highly variable will have a larger variance than one which is relatively homogeneous.

1.6.3 Standard Deviation

The positive square root of the variance, $S = \sqrt{S^2}$.

- If every value in the data set is increased by a constant c: the mean increases by c, but the standard deviation remains unaltered.
- If every value is multiplied by a constant k: the mean is multiplied by k, and the standard deviation is also multiplied by k (more precisely $|k|$).

1.6.4 Skewness

One measure of skewness is **Pearson's Coefficient of Skewness**:

$$\text{Pearson's Coefficient of Skewness} = \frac{\text{Mean} - \text{Mode}}{\text{Standard Deviation}} \approx \frac{3(\text{Mean} - \text{Median})}{\text{Standard Deviation}}$$

Skewness generally takes values between -3 and $+3$.

- A value near 0 indicates an approximately symmetric distribution.
- A positive value indicates **right (positive) skew** (a longer tail to the right).
- A negative value indicates **left (negative) skew** (a longer tail to the left).

1.7 The Empirical Rule

For a symmetrical, bell-shaped (approximately normal) distribution with mean μ and standard deviation σ:

- Approximately **68%** of the data lies within $\pm 1\sigma$ of the mean, i.e. in $(\mu - \sigma,\ \mu + \sigma)$.
- Approximately **95%** of the data lies within $\pm 2\sigma$ of the mean, i.e. in $(\mu - 2\sigma,\ \mu + 2\sigma)$.
- Approximately **99.7%** (sometimes quoted as 99.8%) of the data lies within $\pm 3\sigma$ of the mean, i.e. in $(\mu - 3\sigma,\ \mu + 3\sigma)$.

- The total area under the density curve is always 1 (100%).

Since the curve is symmetric, each "side" of an interval carries half of that interval's total percentage: e.g. exactly 34% of the data lies between μ and $\mu + \sigma$ (half of the 68% within $\pm 1\sigma$), 13.5% lies between $\mu + \sigma$ and $\mu + 2\sigma$ (half of $95\% - 68\% = 27\%$), and 2.5% lies beyond $\mu + 2\sigma$ (half of $100\% - 95\% = 5\%$), and so on – these specific percentages (2.5%, 13.5%, 34%, 47.5%, 50%) are frequently tested directly.

Worked example: Weights of first-year students follow a symmetrical bell-shaped distribution with mean 60 kg and variance 49 (so $\sigma = 7$ kg). Using the empirical rule, the percentage of weights between 60 kg and 67 kg (i.e. between μ and $\mu + \sigma$) is 34%.

2 Introduction to Probability

2.1 Random Experiments, Outcomes, and Sample Space

- **Random Experiment**: an experiment in which there is some uncertainty about the outcome that will be observed.
- **Outcome**: the result of a single observation of the experiment.
- **Outcome Space** (universal set / **sample space**, Ω or S): the set of *all* possible outcomes of the experiment.
- **Equally Likely Outcomes**: if any outcome of a random experiment is as likely to occur as any other outcome, they are called equally likely outcomes.

Example: rolling a die – outcomes $\{1, 2, 3, 4, 5, 6\}$.

2.2 Events

- **Event**: a subset of the outcome space.
- **Equally Likely Events**: if any event is as likely to occur as any other event.
- **Mutually Exclusive Events**: two events A and B are mutually exclusive (or disjoint) if $A \cap B = \varnothing$.
- **Exhaustive Events**: events whose union covers the whole sample space, i.e. $A \cup B = S$ (so $P(A \cup B) = 1$).
- **Elementary Event**: an event with just a single outcome.

2.3 The Probability Function and Its Axioms

A complete specification of a probability model requires:

- the outcome space Ω corresponding to the random experiment;
- the domain G of the probability function (a set of events);
- the probability function P itself.

A probability function P, defined on domain G, assigns to each event $A \in G$ a unique real number $P(A)$ satisfying the following **axioms**:

- **Axiom 1**: $P(A) \geq 0$ for any $A \in G$.
- **Axiom 2**: $P(\Omega) = 1$.

- **Axiom 3**: $P(A \cup B) = P(A) + P(B)$ whenever $P(A \cap B) = \varnothing$ (i.e. whenever A and B are mutually exclusive), for any $A, B \in G$.

2.3.1 Properties of P

- **Property 1**: $P(A') = 1 - P(A)$.
  Proof: $A \cap A' = \varnothing$, so $P(A) + P(A') = P(A \cup A')$ by Axiom 3 $= P(\Omega) = 1$ (since $A \cup A' = \Omega$), by Axiom 2. Hence $P(A') = 1 - P(A)$.
- **Property 2**: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

2.4 Determining Probabilities in Practice

2.4.1 Symmetry Considerations

Physical symmetry sometimes lets us easily assign probability values to events. E.g. for a properly-constructed uniform six-sided die, each of the six elementary outcomes is equally likely, so

$$P(A) = \frac{n(A)}{n(\Omega)}$$

where $n(A)$ is the number of outcomes favourable to A, and $n(\Omega)$ is the total number of possible outcomes.

2.4.2 Probability by Relative Frequency

If a random experiment can be repeated, and $f_r(A)$ denotes the number of times event A occurs in r repetitions of the experiment (the *frequency* of A), then for large r we often assign

$$P(A) \approx \frac{f_r(A)}{r}$$

Example: a drawing pin is dropped 1000 times and lands point-down 721 times; we assign $P(\text{Down}) = 0.721$.

2.5 Worked Examples on Basic Probability

- **Two marbles from a bag** (3 red, 2 blue, drawn simultaneously): using an *extended* (equally-likely) outcome space that individually labels each marble (e.g. R_1, R_2, R_3, B_1, B_2) gives 10 equally likely pairs. $P(\text{1 red and 1 blue}) = 5/10$; $P(\text{at least 1 blue}) = 7/10$.
- **A card from a 52-card deck**: $P(\text{a seven}) = 4/52$; $P(\text{not a seven}) = 1 - 4/52 = 48/52$; $P(\text{a heart}) = 13/52$; $P(\text{not a face card}) = 1 - 12/52 = 40/52$ (there are 12 face cards: J, Q, K in each of 4 suits); $P(\text{number divisible by 3}) = 12/52$ (numbers 3,6,9 in each of 4 suits).
- **A coin and a die thrown together** (12 equally likely outcomes): $P(\text{a head}) = 6/12 = 1/2$; $P(\text{number} > 4) = 4/12$; $P(\text{head and number} > 4) = 2/12$; $P(\text{head or number} > 4) = 8/12$ (using Property 2: $6/12 + 4/12 - 2/12$).

3 Conditional Probability, Total Probability, and Bayes' Theorem

3.1 Conditional Probability

If A and B are two events with $P(A) \neq 0$ and $P(B) \neq 0$, the probability of A *given that* B has already occurred is written $P(A|B)$ ("probability of A, given B"), defined as:

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{for } P(B) \neq 0, \qquad P(B|A) = \frac{P(A \cap B)}{P(A)} \quad \text{for } P(A) \neq 0$$

This is often rearranged as the **multiplication rule**:

$$P(A \cap B) = P(B|A)P(A) = P(A|B)P(B)$$

Clearly $P(B|B) = 1$, since we are certain that B has already occurred.

Worked examples:

- Given a heart is picked at random from a 52-card deck, the probability it is a picture (face) card: let A = picture card, B = heart. There are 3 picture cards among the 13 hearts, so $P(A|B) = \dfrac{P(A \cap B)}{P(B)} = \dfrac{3/52}{13/52} = \dfrac{3}{13}$.
- When a die is thrown, an odd number occurs; probability the number is prime: let A = prime, B = odd. Odd numbers: $\{1, 3, 5\}$; primes that are also odd: $\{3, 5\}$. $P(A|B) = \dfrac{2/6}{3/6} = \dfrac{2}{3}$.
- **Bag with 10 counters (7 green, 3 white), drawn without replacement**: let G_1: first counter green, W_2: second counter white.

$$\begin{aligned}
P(G_1) &= \frac{7}{10} \\
P(W_2|G_1) &= \frac{3}{9} \quad \text{(9 counters remain, 3 white)} \\
P(W_2 \cap G_1) &= P(W_2|G_1)P(G_1) = \frac{3}{9} \times \frac{7}{10} = \frac{7}{30} \\
P(G_2 \cap W_1) &= P(G_2|W_1)P(W_1) = \frac{7}{9} \times \frac{3}{10} = \frac{7}{30} \\
P(\text{different colours}) &= P(W_2 \cap G_1) + P(G_2 \cap W_1) = \frac{7}{30} + \frac{7}{30} = \frac{7}{15}
\end{aligned}$$

3.2 The Law of Total Probability

Observe that for any event B: $B \cup B' = \Omega$ and $B \cap B' = \varnothing$. From a Venn diagram, $A = (A \cap B) \cup (A \cap B')$, and these two pieces are disjoint, so by Axiom 3:

$$P(A) = P(A \cap B) + P(A \cap B')$$

Using the definition of conditional probability, this gives the important identity:

$$P(A) = P(A|B)P(B) + P(A|B')P(B')$$

This is a particular (2-event) case of the general **Law of Total Probability**.

General Law of Total Probability: suppose $B_1, B_2, \ldots, B_n$ are mutually exclusive and exhaustive events, and A is an arbitrary event. Then:

$$P(A) = \sum_{i=1}^{n} P(A \cap B_i) = \sum_{i=1}^{n} P(A|B_i)P(B_i)$$

$$P(A) = P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + \cdots + P(A|B_n)P(B_n)$$

Proof: since $B_1, \ldots, B_n$ are mutually exclusive and exhaustive, the events $A \cap B_1, A \cap B_2, \ldots, A \cap B_n$ are mutually exclusive, and their union is exactly A. By repeated use of Axiom 3, $P(A) = \sum_i P(A \cap B_i) = \sum_i P(A|B_i)P(B_i)$.

Worked example: a bag contains one fair coin, two double-headed coins, and three double-tailed coins. A coin is picked at random and tossed. Let H = lands Heads, C_F, C_H, C_T be the events fair/double-headed/double-tailed coin is selected.

$$P(H) = P(H|C_F)P(C_F) + P(H|C_H)P(C_H) + P(H|C_T)P(C_T) = \tfrac{1}{2} \cdot \tfrac{1}{6} + 1 \cdot \tfrac{2}{6} + 0 \cdot \tfrac{3}{6} = \frac{5}{12}$$

Worked example (car starting): $P(\text{rain}) = 0.4$. If it rained, car starts first time with probability 0.6; if not, with probability 0.9.

$$P(\text{starts}) = P(S|R)P(R) + P(S|R')P(R') = (0.6)(0.4) + (0.9)(0.6) = 0.24 + 0.54 = 0.78$$

3.3 Bayes' Theorem

Suppose $A_1, A_2, \ldots, A_n$ are mutually exclusive and exhaustive events, and B is an arbitrary event with $P(B) \neq 0$. Then for $i = 1, 2, \ldots, n$:

$$P(A_i|B) = \frac{P(B|A_i)P(A_i)}{P(B)} = \frac{P(B|A_i)P(A_i)}{\sum_{j=1}^{n} P(B|A_j)P(A_j)}$$

Proof: follows directly from the definition of conditional probability combined with the Law of Total Probability (used to expand the denominator $P(B)$).

For just two events, this simplifies to:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)} \quad \text{for } P(B) \neq 0$$

3.3.1 Worked Examples

Medical test example: Let A: person has the disease; B: test result is positive. Given $P(B|A) = 0.99$, $P(B|A^c) = 0.005$, and $P(A) = 0.001$ (0.1% prevalence), so $P(A^c) = 0.999$. Find $P(A|B)$:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|A^c)P(A^c)} = \frac{(0.99)(0.001)}{(0.99)(0.001) + (0.005)(0.999)} \approx 0.165$$

This striking result (only about a 16.5% chance of actually having the disease, despite a positive test with 99% sensitivity) illustrates why base-rate/prevalence matters: with a rare disease, false positives from the much larger healthy population can outnumber true positives.

Manufacturing plants example: A company has 3 plants producing 50%, 30%, 20% of output respectively, with defect rates 0.02, 0.05, 0.01 respectively.

a) $P(\text{defective}) = (0.5)(0.02) + (0.3)(0.05) + (0.2)(0.01) = 0.010 + 0.015 + 0.002 = 0.027$ (Law of Total Probability).

b) $P(\text{plant 2} \mid \text{defective}) = \dfrac{(0.3)(0.05)}{0.027} = \dfrac{0.015}{0.027} \approx 0.556$ (Bayes' Theorem).

4 Independence of Events

4.1 Definition

Events A and B are said to be **independent** if one of the events can occur without being affected by the other – i.e. the probability of A occurring is unchanged by knowing whether or not B has occurred. Formally, A and B are independent if:

$$P(A \cap B) = P(A)P(B)$$

4.2 Property 1

If $P(A) > 0$ and $P(B) > 0$, then A and B are independent if and only if

$$P(A|B) = P(A) \qquad \text{and} \qquad P(B|A) = P(B)$$

Proof ($\Rightarrow$): if A, B independent, $P(A \cap B) = P(A)P(B)$. Then $P(A|B) = P(A \cap B)/P(B) = P(A)P(B)/P(B) = P(A)$.
Proof ($\Leftarrow$): if $P(A|B) = P(A)$, then by the conditional probability definition, $P(A \cap B)/P(B) = P(A)$, so $P(A \cap B) = P(A)P(B)$, which is exactly the definition of independence.

4.3 Property 2

If A and B are independent events, then A and B' are also independent events.

4.4 Independence of More Than Two Events

Three events A, B, C are **mutually independent** if and only if *all* of the following hold:

$$P(A \cap B) = P(A)P(B), \quad P(A \cap C) = P(A)P(C), \quad P(B \cap C) = P(B)P(C), \quad \text{and}$$

$$P(A \cap B \cap C) = P(A)P(B)P(C)$$

(Note: it is *not* sufficient for only the pairwise conditions to hold – all four conditions above must be satisfied for mutual independence; events satisfying only the pairwise conditions are called *pairwise independent*, which is weaker.)

Property: if A, B, C are mutually independent, then (among other consequences):

a) A and $B \cup C$ are independent.

b) A and $B \cap C$ are independent.

c) A, B and C' are mutually independent.

d) A, B' and C' are mutually independent.

4.5 Worked Examples

- **Die thrown twice**: A: 4 on first throw ($P(A) = 1/6$); B: odd number on second throw ($P(B) = 1/2$). Since the second throw is physically unaffected by the first, A, B are independent, and indeed $P(A \cap B) = 3/36 = (1/6)(1/2) = P(A)P(B)$.

- **Sampling with replacement**: a bag has 5 red, 7 black counters; a counter is drawn, noted, and *replaced*, then a second is drawn. A: first is red ($P(A) = 5/12$); B: second is black ($P(B) = 7/12$). Because of replacement, A, B are independent, so $P(A \cap B) = (5/12)(7/12) = 35/144$.

- **Given** $P(A) = 1/3$, $P(A \cap B) = 1/12$, A, B independent: find $P(B)$ and $P(A \cup B)$.

$$P(B) = \frac{P(A \cap B)}{P(A)} = \frac{1/12}{1/3} = \frac{1}{4}, \qquad P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{1}{3} + \frac{1}{4} - \frac{1}{12} = \frac{1}{2}$$

- **Fair die thrown twice**: probability neither throw is a 4:

$$P(A' \cap B') = P(A')P(B') = \frac{5}{6} \cdot \frac{5}{6} = \frac{25}{36}$$

  Probability at least one throw is a 4: $1 - \frac{25}{36} = \frac{11}{36}$.

- **Given** $P(A) = 1/4$, $P(A|B) = 1/2$, $P(B|A) = 2/3$:
  a) Since $P(A|B) = 1/2 \neq P(A) = 1/4$, A and B are **not** independent.
  b) $P(A \cap B) = P(B|A)P(A) = \frac{2}{3} \cdot \frac{1}{4} = \frac{1}{6}$.
  c) Since $P(A \cap B) = 1/6 \neq 0$, A and B are **not** mutually exclusive.
  d) Using $P(A|B)P(B) = P(B|A)P(A)$: $\frac{1}{2}P(B) = \frac{2}{3} \cdot \frac{1}{4} \Rightarrow P(B) = \frac{1}{3}$.

4.6 Summary: Mutually Exclusive vs Independent

These are frequently confused but are *different* concepts, and (except in degenerate cases) mutually exclusive events with nonzero probability are **never** independent:

- **Mutually exclusive**: $P(A \cap B) = 0$ (the events cannot both occur).
- **Independent**: $P(A \cap B) = P(A)P(B)$ (occurrence of one does not affect the probability of the other).
- If A, B are mutually exclusive with $P(A) > 0$ and $P(B) > 0$, then $P(A \cap B) = 0 \neq P(A)P(B)$, so they cannot be independent.

5 Discrete Random Variables

5.1 Random Variables

Many real-world experiments produce uncertain outcomes; to analyse them mathematically, we often convert outcomes into numbers. Let Ω be the outcome space of a random experiment. A function $X : \Omega \to \mathbb{R}$ is called a **random variable**.

5.2 Discrete Random Variables

X is a **discrete random variable** if the values it can take form a finite (or countably infinite) set of sample points.

Example: toss a coin 3 times, $\Omega = \{hhh, hht, hth, thh, htt, tht, tth, ttt\}$. Let X = number of Heads observed, Y = number of Tails observed, $Z = X - Y$. Each of X, Y, Z is a random variable (a function from Ω to $\mathbb{R}$), and can be tabulated for every outcome $\omega \in \Omega$.

5.3 Probability Distributions

Let X be a discrete random variable with sample space Ω_X (the set of values X can take). We are interested in the event "X takes the value x", and the probability associated with it.

Definition: the **probability function** of X, denoted P_X (or $p(x)$), is:

$$P_X(x) = \begin{cases} P(X = x) & \text{if } x \in \Omega_X \\ 0 & \text{otherwise} \end{cases}$$

The **probability distribution** of X describes how the total probability mass ($= 1$) is distributed over the values in Ω_X.

5.4 Functions of Random Variables

It is sometimes useful to define new random variables as functions of existing ones – e.g. if X is a random variable, then so are $2X + 5$ and X^2.

Worked example (rectangles): switch A selects length $L \in \{2, 3\}$ cm and switch B selects width $W \in \{1, 2, 3\}$ cm, with given joint probabilities $P(L, W)$ for each of the 6 combinations. From the joint table, we can derive:

- p_L, the (marginal) probability distribution of L: sum the joint probabilities over all W for each value of L.
- p_W, the (marginal) probability distribution of W: sum over all L for each value of W.
- p_X, the probability distribution of the perimeter $X = 2L+2W$: for each (L, W) combination, compute X, then collect/sum probabilities of combinations giving the same X value.

5.5 Expectation

The **expectation** (or **mean**, or **expected value**) of a random variable X defined on outcome space Ω is:

$$E(X) = \sum_{\omega \in \Omega} X(\omega)\, P(\{\omega\})$$

Equivalently, summing over the values of X directly:

$$E(X) = \sum_{x \in \Omega_X} x\, P(x)$$

5.5.1 Properties of Expectation

1. $E(X) = \sum_{x \in \Omega_X} xP(x)$.
2. **Linearity**: $E(aX + b) = aE(X) + b$ for constants a, b.
3. If X and Y are random variables on the same outcome space Ω:

$$E(X + Y) = E(X) + E(Y)$$

5.6 Variance

The mean of X is $E(X) = \mu$. The **variance** of X is:

$$\sigma_X^2 = E[(X - \mu)^2] = \sum_{x \in \Omega_X} (x - \mu)^2 P(x)$$

σ_X (the positive square root) is the **standard deviation** of X. Both variance and standard deviation measure the spread of a probability distribution; the standard deviation is more commonly used since it has the same units as X.

5.6.1 Properties of Variance

1. **Property 1 (computational formula)**: $\mathrm{Var}(X) = E(X^2) - \mu^2$.
2. **Property 2**: for a constant c: $\mathrm{Var}(cX) = c^2\,\mathrm{Var}(X)$.
3. **Property 3**: if X and Y are *independent* random variables:

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) \qquad \text{and} \qquad \mathrm{Var}(X-Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$$

(Note the + sign in *both* formulas – variances always add for independent random variables, even when the variables themselves are subtracted, because variance is a measure of spread, and both sources of variability contribute positively to the total spread.)

5.7 Worked Examples

- **Coin toss for money**: toss an unbiased coin once; win \$k for Heads, lose \$k for Tails. $X \in \{-k, +k\}$ each with probability 0.5.

$$E(X) = (-k)(0.5) + (k)(0.5) = 0, \qquad E(X^2) = k^2(0.5) + k^2(0.5) = k^2$$

$$\mathrm{Var}(X) = E(X^2) - \mu^2 = k^2 - 0 = k^2 \qquad \Rightarrow \qquad \sigma_X = k$$

- **Die game**: win \$20 if a 2 appears, \$40 if a 4 appears, lose \$30 if a 6 appears, otherwise \$0 (each face has probability 1/6).

$$E(X) = 20\left(\frac{1}{6}\right) + 40\left(\frac{1}{6}\right) - 30\left(\frac{1}{6}\right) + 0\left(\frac{3}{6}\right) = \frac{20+40-30}{6} = \frac{30}{6} = 5$$

  The expected winnings are \$5.

- **"Fair game" problems** (as tested in past papers): a game is deemed **fair** if the expected net gain to the player is exactly 0, i.e. $E(\text{payout}) = \text{cost to play}$. To find a required stake/penalty that makes a game fair, set $E(\text{net profit}) = 0$ and solve for the unknown amount.

6 Discrete Probability Distributions

6.1 Discrete Uniform Distribution

Sample space $S = \{1, 2, 3, \ldots, k\}$; each outcome is equally likely. The random variable X defined by $X(i) = i$ has distribution:

$$P(X = x) = \frac{1}{k}, \qquad x = 1, 2, \ldots, k$$

Moments:

$$\text{Mean} = \mu = \frac{1+2+\cdots+k}{k} = \frac{k(k+1)/2}{k} = \frac{k+1}{2}, \qquad \text{Variance} = \sigma^2 = \frac{k^2-1}{12}$$

6.2 Binomial Distribution

Consider an experiment with two possible outcomes: "success" and "failure". A **Binomial situation** arises when n *independent* trials of the experiment are conducted (e.g. tossing a coin 6 times, counting Heads as success; rolling a die 10 times, counting an even number as success).

Let p be the probability of success on any single trial (so $1-p$ is the probability of failure). The probability that the event happens exactly k times in n independent trials is:

$$\Pr(X=k)=\binom{n}{k}p^k(1-p)^{n-k}, \qquad k=0,1,\dots,n$$

When X (the number of successes in n independent trials) follows this distribution, we write $X \sim \mathrm{Bin}(n,p)$; it is called the Binomial distribution because, for $x=0,1,\dots,n$, these probabilities correspond to successive terms of the binomial expansion of $(p+(1-p))^n$.

Example: probability of getting exactly 2 Heads in 6 tosses of a fair coin: $X \sim \mathrm{Bin}(6,1/2)$, $P(X=2)=\binom{6}{2}(1/2)^2(1/2)^4=15/64$.

6.2.1 Properties of the Binomial Distribution

$$\text{Mean}=np, \qquad \text{Variance}=npq \quad \text{where } q=1-p$$

Standard deviation $=\sqrt{npq}$.

6.2.2 Worked Examples

- **3 coin tosses,** $X \sim \mathbf{Bin}(3,1/2)$:
  - $P(3\text{ Heads})=\binom{3}{3}(1/2)^3(1/2)^0=1/8$.
  - $P(2\text{ Tails and 1 Head})$: letting $Y=$ number of Tails, $Y \sim \mathrm{Bin}(3,1/2)$, so $P(Y=2)=\binom{3}{2}(1/2)^2(1/2)^1=3/8$.
  - $P(\text{at least 1 Head})=1-P(0\text{ Heads})=1-(1/2)^3=7/8$.
  - $P(\text{not more than 1 Tail})=P(0\text{ or 1 Tail})=P(3\text{ or 2 Heads})$.
- **5 tosses of a fair die,** $X \sim \mathbf{Bin}(5,1/6)$, probability that a 3 appears: (a) exactly twice: $\binom{5}{2}(1/6)^2(5/6)^3$; (b) at most once: $P(X=0)+P(X=1)$; (c) at least twice: $1-P(X=0)-P(X=1)$.
- **Family of 4 children,** $X \sim \mathbf{Bin}(4,1/2)$ (probability of a boy $=1/2$):
  - $P(\text{at least 1 boy})=1-P(0\text{ boys})=1-(1/2)^4=15/16$.
  - $P(\text{at least 1 boy and at least 1 girl})=1-P(\text{all boys})-P(\text{all girls})=1-(1/2)^4-(1/2)^4=14/16=7/8$.
- **Expected number of Heads in 100 tosses**: $E(X)=np=100\times 0.5=50$.
- **Defective bolts**: probability of a defective bolt is 0.1, sample of 400 bolts. Mean $=np=400(0.1)=40$; standard deviation $=\sqrt{npq}=\sqrt{400(0.1)(0.9)}=6$.
- **Given mean and sd, find** n,p: if $X \sim \mathrm{Bin}(n,p)$ with mean 5.76 and sd 1.92: from $np=5.76$ and $np(1-p)=1.92^2=3.6864$, dividing gives $1-p=3.6864/5.76=0.64 \Rightarrow p=0.36$, then $n=5.76/0.36=16$. Then $P(X=6)$ can be computed directly from $\mathrm{Bin}(16,0.36)$.

6.3 Poisson Distribution

A discrete random variable X having probability function

$$P(X=x)=\frac{e^{-\lambda}\lambda^x}{x!}, \qquad x=0,1,2,3,\dots$$

where λ can be any positive value, is said to follow the **Poisson distribution**, written $X \sim \mathrm{Po}(\lambda)$.

6.3.1 Properties

$$E(X) = \lambda, \qquad \text{Var}(X) = \lambda \qquad \Rightarrow \qquad \text{Standard deviation} = \sqrt{\lambda}$$

6.3.2 Uses of the Poisson Distribution

If an event is randomly scattered in time or space, occurring with mean rate λ in a given interval of time/space, and X is the number of occurrences in that interval, then $X \sim \text{Po}(\lambda)$. Typical applications: number of car accidents on a road in a day; accidents in a factory in a week; telephone calls to a switchboard in a given minute; insurance claims to a company in a given period.

6.3.3 Worked Examples

- $X \sim \text{Po}(2)$: $P(X = 4) = \dfrac{e^{-2}2^4}{4!}$; $P(X \geq 3) = 1 - P(X = 0) - P(X = 1) - P(X = 2)$.
- **Bacteria example**: mean 4 bacteria per ml, $X \sim \text{Po}(4)$. $P(\text{no bacteria}) = e^{-4}$; $P(4 \text{ bacteria}) = \dfrac{e^{-4}4^4}{4!}$; $P(\text{less than } 3) = P(X = 0) + P(X = 1) + P(X = 2)$.
- **Scaling the interval**: if the rate is 4 per ml, then for 3 ml, let $Y \sim \text{Po}(3 \times 4) = \text{Po}(12)$ (the Poisson parameter scales linearly with the size of the interval).

6.3.4 Normal Approximation to the Poisson Distribution

Since $E(X) = \lambda$ and $\text{Var}(X) = \lambda$, for **large** λ we can approximate $X \sim \text{Po}(\lambda)$ by $X \sim N(\lambda, \lambda)$ (a Normal distribution with mean λ and variance λ), applying a **continuity correction** since we are approximating a discrete distribution with a continuous one.

Worked example: radioactive counts follow Po(25) (mean 25/sec). To find $P(23 \leq X \leq 27)$ using the Normal approximation:

$$X \approx N(25, 25) \quad (\sigma = 5), \qquad P(23 \leq X \leq 27) \approx P(22.5 < Y < 27.5) = P\left(\frac{22.5 - 25}{5} < Z < \frac{27.5 - 25}{5}\right)$$

(the ± 0.5 is the continuity correction, since we are approximating the discrete event $\{23, 24, \ldots, 27\}$ by a continuous interval).

6.4 Geometric Distribution

Consider a sequence of i.i.d. Bernoulli trials, each with $P(\text{success}) = p$. Let X be the number of trials *up to and including* the first success. For $X = x$ there must be a run of $(x - 1)$ failures followed by a success, giving:

$$P(X = x) = p(1 - p)^{x-1}, \qquad x = 1, 2, 3, \ldots; \quad 0 < p < 1$$

We write $X \sim \text{Geometric}(p)$.

Moments:

$$\text{Mean} = \mu = \frac{1}{p}, \qquad \text{Variance} = \sigma^2 = \frac{1 - p}{p^2}$$

Worked example: probability of a male/female child is equal ($p = 1/2$); probability that the family's *fourth* child is their first son: $P(X = 4) = (1/2)^3(1/2) = (1/2)^4 = 1/16$.

Worked example (bird watcher): $P(\text{sees eagle on a given day}) = 1/8 = p$. Probability he first sees an eagle on the third day: $P(X = 3) = (7/8)^2(1/8)$.

6.5 Negative Binomial Distribution

A generalisation of the Geometric distribution: let X be the number of trials on which the k**-th** success occurs, where k is a fixed positive integer. Then:

$$P(X = x) = \binom{x-1}{k-1} p^k (1-p)^{x-k}, \qquad x = k, k+1, k+2, \ldots; \quad 0 < p < 1$$

We say X has a **Negative Binomial**(k, p) distribution. The Geometric distribution is the special case $k = 1$.

Moments:

$$\text{Mean} = \mu = \frac{k}{p}, \qquad \text{Variance} = \sigma^2 = \frac{k(1-p)}{p^2}$$

Worked example (oil wells): $P(\text{strike oil}) = 0.2$. Probability the *third* strike comes on the *seventh* well drilled: $k = 3$, $p = 0.2$, $x = 7$:

$$P(X = 7) = \binom{6}{2}(0.2)^3(0.8)^4$$

Worked example (candy bars): 40% chance of selling per house, needs 5 sales; probability the 5th sale happens at the 11th house: $x = 11, k = 5, p = 0.4$:

$$P(X = 11) = \binom{10}{4}(0.4)^5(0.6)^6$$

6.6 Hypergeometric Distribution

The finite-population equivalent of the Binomial distribution: objects are selected *at random, one after another, without replacement*, from a finite population of N objects consisting of k "successes" and $N - k$ "failures". Because there is no replacement, the trials are **not** independent.

Let X be the number of successes in a sample of size n drawn from this population of size N (which has k successes and $N - k$ failures). Then:

$$P(X = x) = \frac{\binom{k}{x}\binom{N-k}{n-x}}{\binom{N}{n}}, \qquad x = 0, 1, 2, \ldots, \min(k, n)$$

Moments:

$$\text{Mean} = \mu = \frac{nk}{N}, \qquad \text{Variance} = \sigma^2 = \frac{nk(N-k)(N-n)}{N^2(N-1)}$$

Worked example: a deck of 20 cards (6 red, 14 black); 5 cards drawn without replacement; probability exactly 4 are red: $N = 20, k = 6, n = 5, x = 4$:

$$P(X = 4) = \frac{\binom{6}{4}\binom{14}{1}}{\binom{20}{5}}$$

Worked example: 60 applicants, 40 male; 20 selected at random; probability exactly 10 are male: $N = 60, k = 40, n = 20, x = 10$:

$$P(X = 10) = \frac{\binom{40}{10}\binom{20}{10}}{\binom{60}{20}}$$

6.7 Choosing the Right Discrete Distribution

A key exam skill is recognising *which* distribution applies to a given scenario:

- **Binomial**: fixed number n of *independent* trials, each with the *same* probability p of success; counting the **number of successes**. (E.g. sampling *with* replacement, or from a very large/effectively-infinite population.)
- **Poisson**: counting the number of events occurring in a fixed interval of time/space, where events occur independently at a constant average rate λ; no natural upper bound on the count. Often used to approximate a Binomial distribution where n is large and p is small (rare events), taking $\lambda \approx np$.
- **Geometric / Negative Binomial**: independent trials with constant success probability p, but now counting the **number of trials needed** to reach the 1st (Geometric) or k-th (Negative Binomial) success, rather than counting successes in a fixed number of trials.
- **Hypergeometric**: sampling *without replacement* from a small, finite population – trials are *not* independent, and the probability of success changes after each draw.

7 Continuous Random Variables

7.1 Definition

Consider a random variable which can take *any* value in an interval. Such a variable is not discrete, because the values in the interval cannot be placed in one-to-one correspondence with the counting numbers.

Definition: a random variable which can take any value in an interval is called a **continuous random variable**.

7.2 Probability Density Function (p.d.f.)

The probability properties of a continuous random variable X are specified by its **probability density function** $f(x)$, which has the properties:

1. $f(x) \geq 0$ for every $x \in \mathbb{R}$.
2. $\displaystyle\int_{-\infty}^{\infty} f(x)\,dx = 1$ (the total area under the curve is 1).

The probability of the event $\{a < X \leq b\}$ is found from the density function by integrating:

$$P(a < X \leq b) = \int_a^b f(x)\,dx$$

That is, probabilities for a continuous random variable are given by **areas under the density curve**. A key consequence: for a continuous random variable, $P(X = a) = 0$ for any single point a (since the integral over a single point is zero), so $P(a \leq X \leq b) = P(a < X \leq b) = P(a \leq X < b) = P(a < X < b)$ – whether endpoints are included does not affect the probability.

Worked example: $f_T(t) = 0$ for $t < 0$, and $f_T(t) = \lambda e^{-\lambda t}$ for $t \geq 0$ (the exponential density). Then $P(1 < T \leq 2) = \int_1^2 \lambda e^{-\lambda t}\, dt$, which can be evaluated and represented as the corresponding area under the curve between $t = 1$ and $t = 2$.

Worked example: $f(x) = kx$ for $0 \leq x \leq 4$.

(a) Find k: require $\int_0^4 kx\, dx = 1 \Rightarrow k\left[\frac{x^2}{2}\right]_0^4 = 1 \Rightarrow 8k = 1 \Rightarrow k = 1/8$.

(b) So $f(x) = x/8$ for $0 \leq x \leq 4$ (a straight line through the origin).

(c) $P(1 \leq X \leq 2.5) = \int_1^{2.5} \frac{x}{8}\, dx = \left[\frac{x^2}{16}\right]_1^{2.5} = \frac{6.25-1}{16} = \frac{5.25}{16}$.

7.3 Expectation and Variance of a Continuous Random Variable

If X is a continuous random variable with p.d.f. $f(x)$:

$$E(X) = \int_{-\infty}^{\infty} x f(x)\, dx$$

$E(X)$ is often denoted μ and called the mean of X. The variance of X is:

$$\mathrm{Var}(X) = E[(X-\mu)^2] = E(X^2) - \mu^2 = \int_{-\infty}^{\infty} x^2 f(x)\, dx - \mu^2$$

Worked example: $f(x) = \dfrac{3x^2}{64}$ for $0 \leq x \leq 4$.

$$E(X) = \int_0^4 x \cdot \frac{3x^2}{64}\, dx = \frac{3}{64}\int_0^4 x^3\, dx = \frac{3}{64}\left[\frac{x^4}{4}\right]_0^4 = \frac{3}{64} \cdot 64 = 3$$

$$E(X^2) = \int_0^4 x^2 \cdot \frac{3x^2}{64}\, dx = \frac{3}{64}\int_0^4 x^4\, dx = \frac{3}{64}\left[\frac{x^5}{5}\right]_0^4 = \frac{3}{64} \cdot \frac{1024}{5} = \frac{48}{5}$$

$$\mathrm{Var}(X) = E(X^2) - \mu^2 = \frac{48}{5} - 9 = \frac{3}{5}$$

7.4 Continuous Uniform Distribution

A continuous random variable X is **uniformly distributed** on the interval $[a, b]$ if it is equally likely to take any value in that interval, with constant density:

$$f(x) = \frac{1}{b-a}, \qquad a \leq x \leq b \qquad \text{(and } f(x) = 0 \text{ otherwise)}$$

We write $X \sim U(a, b)$.

Moments:

$$\text{Mean} = E(X) = \frac{a+b}{2}, \qquad \text{Variance} = \mathrm{Var}(X) = \frac{(b-a)^2}{12}$$

For any sub-interval $[c, d] \subseteq [a, b]$:

$$P(c \leq X \leq d) = \frac{d-c}{b-a}$$

Worked example: X uniform on $[-5, 3]$. Find $P(-3 < X < 1)$:

$$P(-3 < X < 1) = \frac{1-(-3)}{3-(-5)} = \frac{4}{8} = \frac{1}{2}$$

Worked example: variance of a uniform distribution on $(4, 8)$:

$$\mathrm{Var}(X) = \frac{(8-4)^2}{12} = \frac{16}{12} = \frac{4}{3}$$

8 Normal Distribution

8.1 Introduction

The **Normal distribution** is the most important continuous distribution in statistics. Many measured quantities in the natural sciences approximately follow a normal distribution, e.g. heights, masses, ages, examination results.

8.2 Probability Density Function

A continuous random variable X with p.d.f.

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \qquad x \in \mathbb{R}$$

is said to have a **Normal distribution** with mean μ and variance σ^2, written $X \sim N(\mu, \sigma^2)$. Here μ and σ^2 are the two parameters of the distribution. The distribution is **bell-shaped** and **symmetrical about** $x = \mu$.

8.3 Expectation and Variance

If $X \sim N(\mu, \sigma^2)$:

$$E(X) = \mu \qquad \text{and} \qquad \mathrm{Var}(X) = \sigma^2$$

8.4 The Empirical Rule (revisited for the Normal distribution)

For $X \sim N(\mu, \sigma^2)$:

- The total area under the curve is 1.
- Approximately 68% of the distribution lies within $\pm 1\sigma$ of μ.
- Approximately 95% of the distribution lies within $\pm 2\sigma$ of μ.
- Approximately 99.8% (commonly 99.7%) of the distribution lies within $\pm 3\sigma$ of μ.

8.5 The Standard Normal Distribution

Consider the random variable

$$Z = \frac{X - \mu}{\sigma}$$

If X has a normal distribution with mean μ and variance σ^2, then Z has a **standard normal distribution** with mean 0 and variance 1, written $Z \sim N(0, 1)$. This process is called **standardization**, and lets us use a single set of standard normal tables to compute probabilities for *any* normal distribution.

8.5.1 The p.d.f. for Z

The p.d.f. of the standard normal variable Z is denoted $\phi(z)$:

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

8.5.2 The Cumulative Distribution Function

The **cumulative distribution function (c.d.f.)** of Z, denoted $\Phi(z)$, is:

$$\Phi(z) = P(Z \leq z) = \int_{-\infty}^{z} \phi(u)\, du$$

This integral has no closed form and is difficult to evaluate directly, so in practice we refer to **standard normal tables** (giving $\Phi(z)$ for various z), or use statistical software.

8.5.3 Useful Symmetry Properties

Since the standard normal density is symmetric about 0:

$$\Phi(-z) = 1 - \Phi(z), \qquad P(Z > z) = 1 - \Phi(z), \qquad P(-z < Z < z) = 2\Phi(z) - 1$$

These identities are essential for using one-sided tables to answer two-sided or lower-tail questions.

8.6 Standardizing and Solving Normal Probability Problems

To find $P(a < X < b)$ for $X \sim N(\mu, \sigma^2)$: convert to Z-scores and use the standard normal table.

$$P(a < X < b) = P\left(\frac{a-\mu}{\sigma} < Z < \frac{b-\mu}{\sigma}\right)$$

8.6.1 Worked Examples

- **Finding a cut-off value**: given $Z \sim N(0,1)$ and $P(Z < a) = 0.9750$, from tables $a = 1.96$.
- **Two-sided cut-off**: given $Z \sim N(0,1)$ and $P(|Z| < a) = 0.9500$: since $P(|Z| < a) = 2\Phi(a) - 1$, we get $\Phi(a) = 0.975$, so $a = 1.96$.
- **Task completion time**: time to complete a task is $N(40, 4^2)$ (mean 40 min, sd 4 min). For 90 tasks:
  - Expected number taking less than 45 minutes: find $P(X < 45) = P\big(Z < \frac{45-40}{4}\big) = P(Z < 1.25) = \Phi(1.25)$, then multiply by 90.
  - Expected number between 30 and 47 minutes: find $P(30 < X < 47) = P\big(\frac{30-40}{4} < Z < \frac{47-40}{4}\big) = P(-2.5 < Z < 1.75)$, then multiply by 90.
  - Cut-off points for the central 90%: find k such that $P(40 - k < X < 40 + k) = 0.90$, giving $z = 1.645$ (since $\Phi(1.645) \approx 0.95$), so $k = 1.645 \times 4 = 6.58$ minutes; the central 90% lies between $40 - 6.58$ and $40 + 6.58$ minutes.
- **Symmetric interval width (download speed)**: $X \sim N(120, 15^2)$. Find k such that $P(120 - k < X < 120 + k) = 0.90$:

$$2\Phi(k/15) - 1 = 0.90 \Rightarrow \Phi(k/15) = 0.95 \Rightarrow k/15 = 1.645 \Rightarrow k = 24.675$$

8.7 Normal Approximation to the Binomial Distribution

If $X \sim \text{Bin}(n, p)$ and n is large (with np and $n(1-p)$ both reasonably large, a common rule of thumb being both ≥ 5), then X can be approximated by:

$$X \approx N\big(np,\ np(1-p)\big)$$

Because the Binomial is discrete and the Normal is continuous, a **continuity correction** of ± 0.5 is applied when computing probabilities.

Worked example: $X \sim \text{Bin}(200, 0.95)$. Mean $\mu = np = 190$; variance $\sigma^2 = np(1-p) = 200(0.95)(0.05) = 9.5$, so $\sigma = \sqrt{9.5} \approx 3.0822$. To find $P(180 \leq X \leq 195)$:

$$P(180 \leq X \leq 195) \approx P(179.5 < Y < 195.5) = P\left(\frac{179.5 - 190}{3.0822} < Z < \frac{195.5 - 190}{3.0822}\right)$$

$$= P(-3.407 < Z < 1.7845) = \Phi(1.7845) - \Phi(-3.407) \approx 0.9628 - 0.0003 = 0.9625$$

8.8 Normal Approximation to the Poisson Distribution – Worked Example

$X \sim \text{Po}(100)$ (large $\lambda = 100$, so $X \approx N(100, 100)$, $\sigma = 10$). To find $P(90 \leq X \leq 110)$, apply continuity correction:

$$P(90 \leq X \leq 110) \approx P(89.5 < Y < 110.5) = P\left(\frac{89.5 - 100}{10} < Z < \frac{110.5 - 100}{10}\right) = P(-1.05 < Z < 1.05)$$

$$= 2\Phi(1.05) - 1 = 2(0.8531) - 1 = 0.7062$$

8.9 Estimating Counts in a Population

When a random variable models a population characteristic, probabilities can be converted into *expected counts* by multiplying by the population size.

Worked example: pumpkin mass $\sim N(1\text{kg}, 0.15^2)$. In a lorry load of 800 pumpkins, the estimated number with mass greater than 1.13kg is

$$800 \times P(X > 1.13) = 800 \times P\left(Z > \frac{1.13 - 1}{0.15}\right) = 800 \times P(Z > 0.867)$$

9 Sampling Distributions and the Central Limit Theorem

9.1 Sampling Distribution of the Sample Mean

Suppose a random sample of size n is taken from a population (or distribution) with mean μ and variance σ^2. The **sample mean** is

$$\bar{X} = \frac{X_1 + X_2 + \cdots + X_n}{n}$$

$\bar{X}$ is itself a random variable (it varies from sample to sample), and its own probability distribution is called its **sampling distribution**.

Key results (using linearity of expectation and the variance-of-independent-sum property):

$$E(\bar{X}) = \mu, \qquad \text{Var}(\bar{X}) = \frac{\sigma^2}{n}$$

That is, the sample mean is (on average) equal to the population mean, but its variability *decreases* as the sample size n increases (larger samples give more precise estimates of μ).

9.2 Exact Sampling Distribution When the Population Is Normal

If the underlying population itself is normal, $X \sim N(\mu, \sigma^2)$, then *for any sample size* n (even small), the sample mean is **exactly** normally distributed:

$$\bar{X} \sim N\left(\mu,\ \frac{\sigma^2}{n}\right)$$

9.3 The Central Limit Theorem (CLT)

The **Central Limit Theorem** states that, regardless of the shape of the population distribution (as long as it has finite mean μ and finite variance σ^2), the sampling distribution of the sample mean $\bar{X}$ approaches a **normal distribution** as the sample size n becomes large:

$$\bar{X} \sim N\left(\mu,\ \frac{\sigma^2}{n}\right) \qquad \text{for large } n$$

This is extremely powerful because it lets us use normal-distribution methods to compute probabilities involving $\bar{X}$ *even when the original population is not normal* (e.g. Binomial or Poisson populations), provided the sample size is reasonably large.

9.3.1 Worked Examples

- **Vending machine**: dispensed amount has mean 200ml, sd 15ml (population distribution not specified as normal). For a random sample of size $n = 36$, find the probability the sample mean is at least 204ml:

$$\text{By the CLT: } \bar{X} \sim N\left(200, \frac{225}{36}\right) \qquad (\text{sd of } \bar{X} = 15/6 = 2.5)$$

$$P(\bar{X} > 204) = P\left(Z > \frac{204 - 200}{15/6}\right) = P(Z > 1.6) = 1 - \Phi(1.6)$$

- **Plant heights**: $X \sim N(21, 90)$ (population already normal, so the result for $\bar{X}$ is *exact*, not merely approximate). Sample of $n = 10$:

$$\bar{X} \sim N\left(21, \frac{90}{10}\right) = N(21, 9) \quad (\text{sd} = 3)$$

$$P(18 < \bar{X} < 27) = P\left(\frac{18 - 21}{3} < Z < \frac{27 - 21}{3}\right) = P(-1 < Z < 2)$$

- **Estimating sample size from a percentile**: samples of size n are drawn from Po(2.5), and approximately 5% of sample means are less than 2.025. Estimate n: since $X \sim \text{Po}(2.5)$ gives $E(X) = \text{Var}(X) = 2.5$, by the CLT $\bar{X} \sim N(2.5, 2.5/n)$ approximately. We need

$$P(\bar{X} < 2.025) = 0.05 \Rightarrow P\left(Z < \frac{2.025 - 2.5}{\sqrt{2.5/n}}\right) = 0.05$$

  From tables, $P(Z < -1.645) = 0.05$, so

$$\frac{-0.475}{\sqrt{2.5/n}} = -1.645 \Rightarrow \frac{0.475}{\sqrt{2.5/n}} = 1.645 \Rightarrow n \approx 30$$

9.4 Difference of Two Independent Sample Means

If $\bar{X}$ and $\bar{Y}$ are the sample means of two **independent** random samples (sizes n_1, n_2) drawn from $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$ respectively, then:

$$\bar{X} \sim N\left(\mu_1, \frac{\sigma_1^2}{n_1}\right), \qquad \bar{Y} \sim N\left(\mu_2, \frac{\sigma_2^2}{n_2}\right)$$

Since $\bar{X}$ and $\bar{Y}$ are independent, and variances of independent random variables add (even under subtraction, see Section 5):

$$(\bar{X} - \bar{Y}) \sim N\left(\mu_1 - \mu_2,\ \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}\right)$$

Worked example: $\bar{X} \sim N(106, 150/30)$ (from $N(106, 150)$, $n = 30$), $\bar{Y} \sim N(103, 200/50)$ (from $N(103, 200)$, $n = 50$), independent.

$$(\bar{X} - \bar{Y}) \sim N(106 - 103,\ 150/30 + 200/50) = N(3,\ 5 + 4) = N(3, 9) \quad (\text{sd} = 3)$$

$$P(\bar{X} - \bar{Y} > 1.2) = P\left(Z > \frac{1.2 - 3}{3}\right) = P(Z > -0.6) = 1 - \Phi(-0.6)$$

9.5 Summary Table of Common Distributions

Distribution	**Notation**	**p.f. / p.d.f.**	**Mean**	**Variance**
Discrete Uniform	–	$1/k$	$\frac{k+1}{2}$	$\frac{k^2-1}{12}$
Binomial	$\text{Bin}(n, p)$	$\binom{n}{x} p^x (1-p)^{n-x}$	np	$np(1-p)$
Poisson	$\text{Po}(\lambda)$	$\frac{e^{-\lambda}\lambda^x}{x!}$	λ	λ
Geometric	–	$p(1-p)^{x-1}$	$\frac{1}{p}$	$\frac{1-p}{p^2}$
Negative Binomial	–	$\binom{x-1}{k-1} p^k (1-p)^{x-k}$	$\frac{k}{p}$	$\frac{k(1-p)}{p^2}$
Hypergeometric	–	$\frac{\binom{k}{x}\binom{N-k}{n-x}}{\binom{N}{n}}$	$\frac{nk}{N}$	$\frac{nk(N-k)(N-n)}{N^2(N-1)}$
Continuous Uniform	$U(a, b)$	$\frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Normal	$N(\mu, \sigma^2)$	$\frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$	μ	σ^2